Synthetic Cultural Agents as Group-Level Instruments: A Preregistered Evaluation from Anchor to Respondents

Abstract

Cross-national surveys commit to items before learning whether they separate countries; language models offer a cheap first reading. The usual route simulates individuals, then averages. We ask the model only about the group. Six declared signs of country-level survey moments label synthetic preference pairs that train adapters, which read items without naming the country. On trust, they order countries as the survey does on two scenario banks. A prompt stating those signs does too, so signs carry the order. Against a retrospective criterion, a preregistered screen of held-out items gives a weak signal that registered contrasts cannot separate from questionnaire readers. Descriptively, the adapters' order does not detectably agree with respondents', and their response distributions lie farther from the survey than the base model's. That is, the instrument orders countries like a benchmark survey while answering unlike another survey's respondents. It can complement human pretesting. We have not tested pre-piloting.

Keywords: synthetic respondents; large language models; measurement validity; survey item screening; preregistration; cross-national measurement.

1 Introduction

Cross-national survey programs commit to their items long before they learn how those items behave. A question about trust must be translated, pretested and fielded in many countries before a researcher sees whether its country differences follow the construct it was written for. Language models offer a cheap first reading of such items (Argyle et al., 2023; Horton et al., 2023; Buskirk et al., 2026). A reading that flagged in advance which new items separate countries would be useful at exactly this stage. We build one such reading and check, link by link, what it does against a benchmark survey and against the respondents of a second survey.

The usual route to a synthetic reading goes through individuals. Silicon sampling describes a person in the prompt, collects the model's answer, and averages many such persons into a country figure (Argyle et al., 2023). That route needs two things that are hard to supply. First, it needs a model of each simulated person. The errors of those simulated people are correlated, because one model with one set of assumptions produces all of them. Averaging more of them therefore need not cancel the bias. Second, it needs a sampling frame: someone must decide which people to simulate and in what proportions. The record so far is cautionary. Synthetic responses are sensitive to prompt wording, compress the variance of human answers and depend on the order and labelling of answer options (Bisbee et al., 2024; Boelaert et al., 2025; Wang et al., 2025; Dominguez-Olmedo et al., 2024). Alignment procedures change how faithfully models simulate attitudes (Lyman et al., 2025), and methodological work in this journal concludes that validation against human data remains essential (Kozlowski and Evans, 2025; Broska et al., 2025). Most empirical work asks whether a model reproduces answers that people already gave, whether it is prompted (Santurkar et al., 2023; Durmus et al., 2024) or fine-tuned on survey response distributions (Cao

et al., 2025; Suh et al., 2025). In the fine-tuned case the human survey sits inside the training signal. Language-model pretesting has also been studied as flaw detection in question wording (Sturgis et al., 2026). Whether a method can say which new items will separate countries in an intended direction is a different test. It needs a decision rule fixed before the human criterion is read.

We take a route that never models individuals. Cross-national research usually wants a country-level quantity, such as how countries rank on generalized trust. A probability survey such as the Global Preferences Survey (GPS; Falk et al., 2018) has already aggregated people into such quantities, with sampling weights. We take its country-level moments as declared inputs and ask the model only for a group-level reading of new item text. In other words, the survey aggregates people, and the language model reads items at the level of the group. An order of countries can survive this design without a representative respondent. It requires only that the instrument's score rise with the group's latent preference. The design has a price. A single choice policy per country has no spread within the country built in, so it can carry an order and has no reason to match the distribution of respondents' answers. This is the representative-agent problem (Kirman, 1992), and it gives a reason to expect that order and distribution can come apart. Because country-level agreement says nothing about individuals (Robinson, 1950), we chose the country as the unit and keep our claims there.

Modelling the group directly is not new on its own. SubPOP (Suh et al., 2025) and Group Preference Optimization (Zhao et al., 2023) predict group-level responses without simulating individuals, but they train on human response distributions and name the group. What is specific here is the combination of three features, and each is a choice a researcher can inspect. First, the inputs are declared aggregate moments. The researcher states which six GPS moments enter, and a sign rule turns them into

labels, so nothing about the conditioning hides in a country name or in human answers. Second, the training data are synthetic preference pairs. Each country's sign profile labels one shared bank of generated response pairs. Direct Preference Optimization (Rafailov et al., 2023), which we read as a regularized Bradley–Terry pairwise choice model, then fits one adapter per profile. No human survey answer enters training. Therefore, a human survey is a criterion the method has never seen, which is not true of models fitted to response distributions. Third, the group is unnamed at evaluation. Items are scored with prompts that never mention a country, so whatever the adapters carry comes from the conditioning. A country-token channel is closed by construction.

We evaluate the construction along a chain of three links, treating each as a separate measurement question in the way Wallach et al. (2025) propose for generative systems. The anchor is the GPS, the instrument is the set of adapters, and the respondents are the people who answered Wave 7 of the World Values Survey (WVS). Prompted models have been compared with the GPS across countries and found to be weak proxies for it (Chen et al., 2026). We built the construction on a purposive development panel of 16 countries. We then evaluate it on all 42 countries covered by both the GPS and Wave 7, with a second set of adapters for all 64 sign profiles on a newly generated scenario bank. At the first link we ask whether training installs the synthetic labels, whether the order of countries survives the new bank, and whether a prompt does as well. A factorial over the 64 sign profiles asks whether the six declared dimensions behave as separate inputs. At the second link we ran a preregistered screen of 223 Wave 7 items that no adapter was built for. A probe that reads only the adapters' scores assigns each item a dimension and a direction, and a fixed rule selects the strongest assignments before any human value is read. That criterion is retrospective, because the survey existed before the probe. Finally, descriptive

analyses added after the registered results had been read set the adapters beside the respondents' composites and response distributions.

What we found on the first link is an order of countries that the declared signs carry. Training installs each country's synthetic labels on scenarios the adapters never saw. On trust, the adapters keep the GPS order of countries when the scenario bank is replaced, whereas a prompt that names the country does not. A prompt that states the same six signs keeps the trust order as well, and the contrast between the adapters and that prompt is unresolved. That is, the signs carry the order, and the evidence gives the trained weights no separate credit. The order is specific to trust. Each declared sign moves its own surface in the intended direction and also moves the others, often by more. Trust is the one surface where its own sign dominates.

Checked against people, the instrument answers unlike them. The retrospective screen gives a weak signal. The registered null is rejected, yet the signal is a small fraction of what two human survey waves achieve on the same items, and its country-bootstrap interval reaches zero. Registered contrasts do not distinguish the adapters from careful readers of the questionnaire on the few items they share, and no dimension passes both the second-bank check and the screen. Descriptively, the adapters' order of countries does not detectably agree with the respondents' composite on any surface. Notably, the GPS's own agreement with those respondents is loose. On response distributions the ranking reverses. The adapters sit farther from the survey's response distributions than the base model does, whereas a country-named prompt sits closer and has no trust order. The separation of order from distributional fidelity is known (da Silva et al., 2026; Wang, 2026). What this case adds is one method, on one anchor, that answers the two questions in opposite directions. Put plainly, a model asked only about the group can order countries the way a cross-national survey does while answering unlike the people in them.

Our contributions are two.

1. **An inspectable group-level conditioning method.** Declared aggregate moments, synthetic preference pairs and an unnamed group at evaluation define a construction whose information budget is on the record. A reader can see what the method was given and check where its readings depart from people.

2. **A reusable evaluation protocol.** We separate the three links and state what a favorable or unfavorable result at each permits. The protocol includes a screen of new items whose probe is fixed before any human value is read, and a permutation null that absorbs the probe's selection rate. It adds a human ceiling from two survey waves, item-matched comparators that read only the questionnaire, and a factorial that tests whether declared dimensions stay separate.

What the paper shows is bounded by its design: one anchor, one instrument family and one base model. We do not show that trained weights improve on a prompt that states the same signs. Nor do we show that the adapters agree with respondents beyond the weak retrospective signal, or that screening would change which items a researcher fields. Pre-piloting is a prospective use that remains untested. The instrument carries no information about magnitudes or about variation among people within a country, and its readings concern differences among countries. Within these bounds a survey designer can consult it beside human pretesting as a complement, never in its place.

Section 2 places the method among current uses of synthetic respondents and in the measurement literature on cross-national comparability. Sections 3 and 4 describe the construction and the design with its registration, Section 5 reports the results, and Section 6 discusses what they permit.

2 Synthetic Respondents as Measurement Instruments

Synthetic respondents and the criterion they need. A language model conditioned on a description of a population can be asked to answer for a sample of its members. In silicon sampling the description enters the prompt, and the model's response distribution is read as an estimate of the population's (Argyle et al., 2023). In the homo silicus program the model plays an economic agent in a stylized decision problem (Horton et al., 2023). A systematic review maps how fast these uses have spread across survey research (Buskirk et al., 2026). Fine-tuning on survey responses is one response to the evidence on prompted models (Suh et al., 2025; Cao et al., 2025; Krsteski et al., 2026). Two reviews in this journal draw the conclusion that matters here: validation against human data remains essential (Kozlowski and Evans, 2025), and model predictions are best treated as potentially informative observations while human subjects remain the gold standard (Broska et al., 2025). In other words, both require a criterion against which the synthetic reading can fail. Our evaluation supplies one criterion for each link of the chain.

Construct, measure, and anchor. Three objects have to be kept apart. A construct is a theoretical preference attribute, such as generalized trust. A measure operationalizes it through observed answers, such as the response distribution to a WVS question about whether most people can be trusted. An anchor is an aggregate score that conditions the instrument, such as a country's GPS trust coordinate. The GPS supplies anchors for six economic preferences: trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity. It measures them with experimentally validated modules in a common design across countries (Falk et al., 2018). We treat

these coordinates as declared aggregate moments: the researcher chooses them, and nothing in the construction certifies them as the right moments for a given survey. Mapping a WVS item to a GPS dimension makes the item a candidate indicator, and an average of such items a proposed composite. Agreement between instrument and anchor, between instrument and people, and between anchor and people are three different facts. They are the three comparisons that the chain makes. The multitrait–multimethod logic of Campbell and Fiske (1959) applies directly. That is, the human survey is the other method, and an instrument that agrees with its own anchor has passed a construction check without yet meeting a criterion.

Cross-national comparability. We chose the country as the unit of analysis. Agreement across countries says nothing on its own about individuals, the problem of ecological inference (Robinson, 1950). Country-level agreement does not establish that an item measures the same construct for individuals in different countries, because loadings and intercepts can differ across groups in ways that leave country means comparable for some purposes and misleading for others (Steenkamp and Baumgartner, 1998; Davidov et al., 2014). Work in this journal argues that country-level composites can suffice for simple group comparisons despite noninvariance, while understating uncertainty (Bratt, 2025). A simulation study in the same journal finds that unaddressed measurement error and, less so, noninvariance bias multilevel estimates (Pokropek et al., 2025). Every comparison with respondents below concerns country rankings, with no claim about invariance or individual-level validity.

3 The Instrument

The instrument is the middle link of the chain. It maps declared aggregate preference signs from the anchor into choice policies and reads them on survey items without naming any country. A language model is treated throughout as a conditional distribution over response sequences given a prompt (Ludwig et al., 2026).

3.1 From declared signs to a shared pair bank

Every input to the instrument should be declared and inspectable. Let $z_c = (z_{c1}, \ldots, z_{cK})$ be the GPS coordinates of country c on the K dimensions of trust, risk taking, patience, altruism, positive reciprocity and negative reciprocity. Only the sign profile s_c, with $s_{ck} = 1\{z_{ck} \geq 0\}$, enters the construction. A teacher model and a response generator produce a shared bank of scenarios, each with two responses intended to express a higher and a lower level of one dimension k_i. For country c, the chosen and rejected responses on pair i are

$$(y_{w,ci},\, y_{\ell,ci}) = \begin{cases} (y_i^+,\, y_i^-), & s_{c,k_i} = 1, \\ (y_i^-,\, y_i^+), & s_{c,k_i} = 0. \end{cases} \tag{1}$$

The scenario and response text is identical for every country. Only the label changes. No scenario names a country or reuses WVS question wording. The intended loading of each pair is a generator judgment and has not been verified as a trait manipulation. Two countries with the same sign profile receive identical labels, so the instrument cannot distinguish them, and nothing in the construction carries the magnitude $|z_{ck}|$. That is the information budget: six bits per country.

3.2 Preference optimization as regularized pairwise choice

Labeled pairs are choice data, so we fit each sign profile with a pairwise choice model, Direct Preference Optimization. Under the Bradley–Terry model, the probability that y_w is chosen over y_ℓ after prompt x is $\sigma\{r(x, y_w) - r(x, y_\ell)\}$, with σ the logistic function and r a latent utility index. This is the binary logit of random-utility theory (Bradley and Terry, 1952). DPO writes the index as $r(x, y) = \beta \log\{\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)\}$ up to a prompt-specific constant that cancels from pairwise differences. It then minimizes the negative log likelihood of the labeled pairs,

$$\mathcal{L}_c(\theta) = -\frac{1}{|\mathcal{D}_c|} \sum_{(x, y_w, y_\ell) \in \mathcal{D}_c} \log \sigma \left\{ \beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)} \right\}. \tag{2}$$

The parameter β penalizes departures from the reference policy, which makes the estimator a regularized Bradley–Terry fit (Rafailov et al., 2023). Because the labels are deterministic, the likelihood is a fitting criterion and says nothing about noise in human choices. Each adapter is a low-rank update to a quantized Llama 3.1 8B Instruct base model (Hu et al., 2022; Dettmers et al., 2023), trained jointly on all six dimensions for one epoch on 526 of the 658 pairs, with LoRA rank 16 and $\beta = 0.1$. A joint adapter does not isolate any single dimension. Whether the dimensions separate in practice is the subject of the factorial in Section 4.

3.3 Two cohorts and two prompt routes

Two kinds of question need two sets of adapters: how countries compare and how each sign acts. The first cohort has one adapter for each of the 42 countries in the intersection of the GPS and WVS Wave 7, trained on the original scenario bank. These countries realize 26 distinct sign profiles. The second cohort is trained on a

second, independently generated bank of 658 pairs under the same sign rule and text restrictions. It contains one adapter for each of the 64 possible sign profiles, a full 2^6 factorial in the six sign bits. The labels depend on the country only through its sign profile. Therefore, each country takes its profile's adapter in country-level comparisons. Most of the 64 profiles are design cells that no country realizes.

Adapters are one channel for the signs, so we also score two prompt routes. The sign-profile prompt route gives the base model the same six signs in words, with no country name and no adapter. It carries the same declared information through a different channel and is the natural comparison for the trained route. The country-named persona route replaces the neutral opening of the evaluation prompt with "Answer this questionnaire as a typical adult living in" followed by the country's name, and supplies no signs. It changes the information as well as the channel, because the model can draw on whatever it associates with the country name. Naming the population is the comparison a survey researcher would reach first, so we score it on all 42 countries.

3.4 Readout

Each item needs one number, so we turn the model's code probabilities into a score. For an item q with candidate response codes j, the instrument computes the teacher-forced log probability of each code after the rendered item prompt and normalizes over the supplied codes,

$$\widehat{P}_u(j \mid q) = \frac{\exp \ell_{uqj}}{\sum_h \exp \ell_{uqh}}, \qquad y_{uq} = \sum_j g_q(j)\, \widehat{P}_u(j \mid q), \tag{3}$$

where ℓ_{uqj} is the summed token log probability of code j for unit u (an adapter or a prompted profile) and g_q is the item's recode. That is, the score y_{uq} is the

expected recoded response. Apart from the persona route, no evaluation prompt names a country. The prompt template, option order and recode are frozen in a scoring specification whose hash every run must reproduce before a model is loaded. On the original 23-item WVS battery, we average item scores into seven surfaces. Five follow trust, risk taking, altruism and the two reciprocities. Patience contributes two surfaces, one for each of its two items. We call this the strict battery. The normalized code probabilities are a scoring convention, and their dispersion need not match any human population's response distributions.

4 Evaluation Design

The evaluation follows the chain, and each comparison permits one kind of statement. Table 1 lists the comparisons. Every statistic carries the label of the surface it was computed on, and we never pool statistics from different surfaces.

4.1 Second-bank check

An order tied to one scenario text would say little about the signs. For each surface of the strict battery we compute the Spearman correlation between the adapters' surface composites and the GPS coordinate of the surface's own dimension over the 42 countries. We do this first for the first cohort (S1) and then for the second cohort, with each country assigned its profile's adapter (S2). Intervals come from a profile-block bootstrap that resamples the 26 realized sign profiles, with 2,000 draws, seed 20261011 and percentile limits. The imprint set is the 4 dimensions whose first-cohort interval excluded zero: trust, risk taking, positive reciprocity and altruism. Patience and negative reciprocity were flat and are reported without interpretation. The rule was fixed before the second cohort was trained: for an imprint dimension whose second-

Table 1: Evaluation surfaces. Every statistic belongs to one, and the status column says how its analysis was fixed.

Label	Units, items and comparison	Status
S1	First-cohort country adapters, strict battery: adapter–GPS rank correlation (model side only)	Basis of the second-bank rule
S2	Second-cohort profile adapters: same statistic as S1	Rule fixed before training
H	Second-cohort adapters on held-out Wave 7 items: signed agreement of selected items with human country means	Preregistered; retrospective criterion
F	Second-cohort adapters and prompted profiles: main effects of six sign bits on seven surfaces (model side only)	V5 planned; rest exploratory
R, N	One unseeded retrain; two near-zero refits (model side only)	Predeclared checks
P	Country-named prompt on the base model: persona–GPS rank correlation (model side only)	One contrast planned; rest descriptive
D	Adapters, factorial and prompt routes: agreement with the GPS and the Wave 7 composite, distance to survey distributions, decomposition of main effects, label recovery	Descriptive, not preregistered

bank interval includes zero, we write no sentence about carry-over to the second bank. We also report the agreement between the two banks over the 26 shared profiles. It measures how much the scenario text changes the instrument and involves no human data.

4.2 The country-named prompt

This route tests whether a country name alone carries the order. We scored the persona route on the strict battery for each of the 42 countries in one run on the frozen stack. We compute the persona–GPS correlation for each surface (P) and one planned contrast, fixed in a design note committed before any persona was scored. That contrast is the first cohort's trust correlation minus the persona route's, recomputed on every draw of the profile-block bootstrap above. All other differences are descriptive.

4.3 The held-out screen

Item pool. The screen tests new items, so its design was fixed before any human value was read. The pool was built from WVS Wave 7 metadata alone: ordinal questions with two to eleven contiguous response codes outside the battery used to develop the adapters. A text review excluded items with non-monotone codes or factual content and repaired wording that did not extract. That leaves 223 items that no adapter was built for (online supplement).

Probe. All 64 second-cohort adapters scored every pool item under the frozen specification. For each item i, the probe regresses the 64 adapter scores on an intercept, the six sign bits coded ± 0.5 and their 15 two-way interactions. That leaves 42 error degrees of freedom. Let t_d be the main-effect t statistic of dimension d.

The item's assigned dimension is $d_i = \arg\max_d |t_d|$, or none when the largest $|t_d|$ is below 2.0, and its sign is $s_i = \text{sign}(b_{d_i})$. The primary rule keeps items whose assigned dimension is in the imprint set and whose $|t|$ is at least 2. It keeps at most the top 25 percent of all pool items by $|t|$, which caps the selection at 56 items. The secondary rule keeps every imprint item above the threshold and is a reliability check only. The probe reads model scores only, and the permutation null absorbs the rate at which noise items cross the threshold.

Statistic and criterion. For each selected item, the signed item agreement is

$$a_i = s_i \cdot \rho\big(\bar{h}_{ci},\, z_{c,d_i}\big), \qquad A = \frac{1}{|S|}\sum_{i \in S} a_i, \tag{4}$$

where ρ is the Spearman correlation over the 42 countries and S is the selected set. The term $\bar{h}_{ci}$ is the survey-weighted Wave 7 mean of item i in country c, with negative codes set to missing. An item–country cell that was never fielded enters as missing and is not imputed. That is, A is the mean signed agreement with human country differences. It is positive when those differences run in the direction the instrument assigned, against the GPS coordinate of the assigned dimension.

Inference. Selecting the strongest items could make A positive by chance, so the registered test is a permutation null over items. Each draw takes as many items from the pool as were selected, gives them the selected items' dimension and sign pairs in random order, and recomputes A. We used 10,000 draws and seed 20261015. The one-sided p is the share of null draws at or above the observed A. A 95 percent interval for A comes from 2,000 bootstrap resamples of the 42 countries. The readings were fixed in advance. If A is not above the null at one-sided $\alpha = 0.05$, the adapters do not

screen better than random assignment. If A is above the null but below one third of the human ceiling, a screening signal exists but is weak.

Human ceiling. For each selected item fielded in both WVS waves 6 and 7 we correlate the two waves' country means over the 28 countries with both waves (World Values Survey Association, 2022). Averaging over items gives the agreement two human surveys reach on these items. The ratio of A to it is a benchmark only.

Per-dimension readings. The registered analysis also reports A_d, the mean of a_i over the selected items assigned to dimension d, with an interval from the same country resamples. These intervals are unadjusted. Therefore, we add exploratory Bonferroni intervals for the four imprint dimensions from the same draws at the 98.75 percent level.

Comparators. A positive A could reflect item wording alone. Two comparators therefore read only the questionnaire and saw no model score or human response. The first is a face-valid rule that assigns an item to the one dimension whose construct words its wording contains. The second is a typed decision model (jev-1.13.0) run 5 times, which assigns an item when at least 3 runs agree on dimension and sign (online supplement). The registered primary contrast is coverage-matched: $D_{\text{common}} = A_{\text{adapter}} - A_{\text{comparator}}$ on the items both assign, with an interval from the paired country bootstrap. An interval above zero would mean the adapters find structure the wording does not reveal. One that covers zero was registered as "the adapters match a careful reader", and one below zero would favor the reader. We report the contrast on each route's own items beside it. The preregistration named in advance that both comparators concentrate on trust.

Readout variants and the prompt route. A selection that moved with option order or phrasing would reflect the prompt. The probe was therefore re-estimated on three re-scorings of the pool (options reversed, two prompt paraphrases). The selection was declared dependent on wording if more than half of the selected items changed dimension or sign. The same probe and test were applied to the 64 prompted profiles.

4.4 Registration, blinding and amendments

The screen was registered in a document committed to the project repository and timestamped by a pushed git tag. That happened before any held-out item was scored by an adapter, when no selection existed and no held-out human value had been read. A freeze guard enforced the order of work (a timeline is in the online supplement). We scored the held-out items, ran the probe and froze the hashes of the selection and the item pool. Only then did we join the human data.

Two amendments were made, both before any human held-out value was read. Amendment 1 repaired 12 items whose standalone wording could not be answered as written, under the pool rule and override mechanism already registered. Every route was then re-scored as one run. We disclose that the first probe pass had been computed and read before the repairs were chosen, and that two repaired items (Q153 and Q157) were in its top quartile. The comparators had assigned the pool on the pre-repair wording. Therefore, Amendment 2 removed the repaired items from both comparator routes before any comparator statistic was computed. It affected one item (Q82), which is outside the adapter selection. That decision preceded any human value. The amendment text, written after the primary statistic was read, records it.

The criterion is retrospective. The Wave 7 file is public and existed before the probe, the pool and the adapters. It is the only human source in this study. Registration guards against tuning the probe to the criterion. It cannot make the criterion prospective, so results on surface H are labelled retrospective throughout. The online supplement discloses the provenance gaps, none of which touches a statistic reported here.

4.5 The sign-profile factorial

The factorial asks whether the six declared signs act as separate inputs. The second cohort covers all 64 sign profiles, so the strict-battery composites form a full factorial in six binary factors. For each of the seven surfaces we fit the probe's regression. That is, a main effect is the difference between the mean composites of profiles with the bit set and without it. Because the bits were assigned by design, this is a causal statement about the instrument. It carries no implication about how preferences operate in people. Each bit has an own surface (patience owns two). The analysis plan set one acceptance check, separate from the held-out registration. At least 4 of the six bits must move their own surface more than any other (V5), against a binomial reference with a one-in-seven chance of a hit. Exploratory views divide effects by each surface's standard deviation over the 64 profiles and count the surfaces whose largest mover is their own bit. They also report the first principal component's variance share and the mean absolute correlation between surfaces. The same analyses on the 64 prompted profiles contrast two instruments that carry the same declared information.

4.6 Retraining noise and the near-zero margin

Effects can only be read against the noise of the pipeline itself. We retrained one profile adapter without a fixed seed, which gives a noise floor. A predeclared sensitivity check also removes the training pairs of every dimension whose GPS coordinate lies within 0.10 of zero. It refits two countries (Great Britain and Colombia), each beside a size-matched control. The rule was to report the margin and never to promote it.

4.7 Descriptive follow-up analyses

Four further analyses were added after the registered results had been read. Two of them are agreement in three columns and distance to the survey's response distributions. The other two are a decomposition of the factorial's cross effects and held-out label recovery, both defined with their results. None is preregistered or tests a rule fixed in advance. We label their numbers D and read them as descriptions. Their human input is the Wave 7 file used for the screen, so every comparison with respondents is retrospective.

Three columns. Agreement with the anchor and with people are different facts, so we compute both. For each of the seven surfaces we compute three Spearman correlations over countries. The first is the adapter composite with the GPS coordinate of the surface's own dimension. The second is the adapter composite with the Wave 7 composite of the same items, and the third is that human composite with the GPS. The adapter columns use the first-cohort adapters with the profile-block bootstrap above (the second cohort is in the supplement). The human–GPS interval resamples countries (seed 20260905). The human columns of risk taking cover one country fewer, which has no valid answer to the risk item.

Distance to the survey distributions. Order says nothing about how close the model's answers are to people's. For country c and item q, the total variation distance between the model's option probabilities and the survey-weighted human answer shares is

$$\mathrm{TVD}_{cq} = \tfrac{1}{2} \sum_{j} \left| \widehat{P}_c(j \mid x_q) - P_c^H(j \mid q) \right|,$$

where j runs over the answer codes and P_c^H is the share of valid human answers equal to j. The headline statistic is the mean over items within a country and then over countries. We score four cells on the same frozen stack (gates in the supplement): the base model and the first-cohort adapters, each with and without the country-named persona prompt. Contrasts are paired differences of country means, each with a profile-block and a country interval (2,000 draws each). The interaction contrast is the adapters' effect with the persona prompt minus their effect without it. In other words, a zero interaction means the two interventions add. The human answer shares enter the distances and are not reported.

5 Results

We report the results along the chain, and each statistic belongs to one surface of Table 1. Everything computed against the Wave 7 file is retrospective. Exact values are in the text, Table 2 and the online supplement.

5.1 Anchor to instrument: what training installs

Every training pair of a country carries a label. The chosen completion is the one that goes with the sign of the country's GPS coordinate on the pair's own dimension. The first question is whether that installation carries over to pairs the adapters never saw.

This is a construction check on the synthetic labels (descriptive, no human data). We split the training pairs once into a training part and a held-out part that the adapters never saw. Recovery is the share of held-out pairs on which the adapter favors the chosen completion more than the base model does, and accuracy is the share on which it prefers that completion outright. Definitions and gates are in the online supplement.

Recovery is 0.940 [0.926, 0.952] and accuracy is 0.816 [0.771, 0.856]. The base model's accuracy is 0.489 [0.471, 0.506]. That interval includes one half, so the base model shows no detectable preference for the chosen completions before training. The mean margin shift is 40.3 [28.2, 54.1] nats. Training installs each country's synthetic labels on scenarios the adapter never saw, which is construction fidelity. It says nothing about whether the labels order countries like the GPS, and nothing about respondents. Notably, the distance results below show that the same adapters lie farther from the survey's response distributions than the base model does.

5.2 Anchor to instrument: the trust order and what carries it

The first cohort encodes four of the six declared signs on the strict battery (S1). Over the 42 countries, the adapter–GPS Spearman correlation is 0.744 [0.500, 0.865] for trust and 0.435 [0.085, 0.686] for risk taking. It is 0.488 [0.120, 0.695] for positive reciprocity and 0.406 [0.020, 0.676] for altruism. These four form the imprint set that the screen later uses.

The second cohort tests whether the order survives a change of scenario text (S2). Trust keeps its agreement with the GPS, at 0.803 [0.615, 0.876]. The other three imprint dimensions keep positive point estimates, but their intervals now include zero (Table 2). Therefore, 1 of the 4 imprint dimensions passes the second-bank check

and 3 do not. Under the rule fixed before the second cohort was trained, we write no sentence about the other three carrying over. Trust is the dimension on which retention of the anchor's order is replicated across scenario banks. The banks also agree profile by profile. Over the 26 shared profiles, the rank correlation between first-bank and second-bank trust composites is 0.907 [0.774, 0.957]. Risk taking is the least stable surface, at 0.544 [0.187, 0.772], which fits its loss on the second bank (the online supplement gives every surface). This agreement measures how much the scenario text changes the instrument and says nothing about people.

What carries the trust order? The comparison a survey researcher would reach for first is a prompt that names the population. It does not reproduce the order (P): the persona–GPS correlation over the 42 countries is 0.028 [−0.339, 0.381]. The planned contrast with the first cohort is 0.716 [0.191, 1.154], and the reading is the same under a country bootstrap. A prompt that states the same six signs in words, with no adapter and no country name, tracks trust as closely as the adapters do. It reaches 0.773 [0.584, 0.844] (panel A of Figure 4, which sets all routes side by side). That is, on trust what separates the routes is whether the declared sign is supplied, whichever channel carries it. The contrast sets declared signs against a country name and says nothing about trained weights against prompting. On the other surfaces the persona route's intervals include zero, except for patience, which it orders positively on one item and negatively on the other (online supplement).

5.3 Inside the instrument: the sign-profile factorial

The factorial asks whether the six declared signs behave as separate inputs (F). On the trained route, every sign bit moves its own surface in the intended direction (panel A of Figure 1; the online supplement gives each effect with its t statistic). All seven own

effects are positive, from 0.364 for trust ($t = 16.74$) to 0.124 for positive reciprocity ($t = 3.02$), and 0 of 7 point the wrong way.

The planned separability check fails. Only 2 of the 6 bits move their own surface more than any other surface, where at least 4 were required (binomial $p = 0.207$). They are the bits for altruism and negative reciprocity. Scale-free views agree (online supplement). Each bit moves its own surface and also moves others, often by more. The risk-taking bit moves the altruism surface by 0.983, against the altruism bit's own effect of 0.242. The trust bit is the largest mover on 3 of the 7 surfaces. One common direction carries 51.3% of the variance of the seven standardized surfaces, and the mean absolute correlation between surfaces is 0.395. In other words, each sign keeps its intended direction on its own surface, while the size of its effects spreads across the others.

The prompted route fails the check by the same count, 2 of 6 (binomial $p = 0.207$), so V5 does not separate the routes. What separates them is the common direction and the sign of the own effects. Under prompting the first principal component carries 83.5% of the variance, the mean absolute correlation between surfaces is 0.807, and the trust bit is the largest mover on all 7 surfaces. Three prompted own effects point against the intended direction. Two are risk taking at -0.457 ($t = -22.31$) and patience Q43 at -0.069 ($t = -19.75$). The third, negative reciprocity at -0.023, is not distinguishable from zero at $t = -1.80$. These are behavioral cross effects of the instrument. The contrast concerns entanglement and the direction of own effects and is silent on screening accuracy, where the difference between routes is unresolved.

We next ask why the trust order survives when the other surfaces lose theirs. We split the second-cohort adapters' main effects into two parts: the effect of each surface's own bit and the summed effect of the other five. We evaluate each part at the sign profile of each of the 42 countries and correlate it with the GPS coordinate

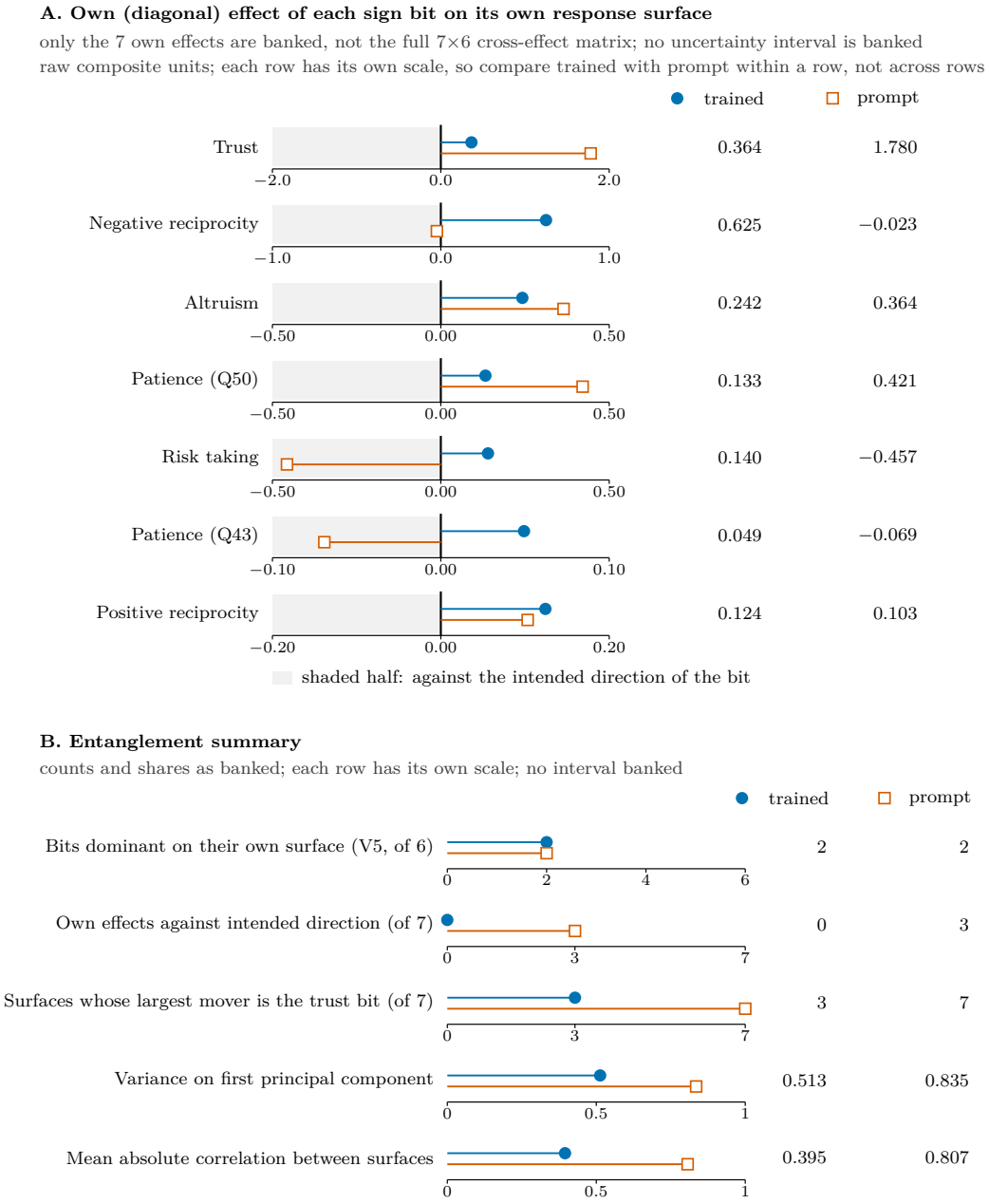


Figure 1: Own effects of the sign bits in the sixty-four-profile factorial (surface F; model side only). (A) For each of seven strict-battery surfaces, the main effect of the surface's own sign bit, for the trained second-cohort adapters (circles) and the sign-profile prompt (squares). All 7 trained own point estimates have the intended sign, and 3 prompted ones do not. (B) Counts and shares that summarize entanglement. V5 was a planned check; the other rows are exploratory.

of the surface's own dimension (D; the online supplement gives every surface). This is a behavioral decomposition that uses model quantities and published GPS values only. It says nothing about what happens inside the network. The own-bit part orders the countries like the GPS on all seven surfaces, with correlations from 0.817 to 0.866. That ordering is mechanical. For a positive own effect, as all seven are, the own part is proportional to the sign of the GPS coordinate. It therefore restates the correlation between signs and GPS and is no separate test of learning. The full prediction, which adds the cross bits, keeps the GPS order on trust at 0.834 and loses most of it elsewhere. On the other surfaces its correlations run from -0.020 for patience Q43 to 0.422 for altruism. The cross part carries little of the full prediction's variation on trust and most of it elsewhere. Var(cross)/Var(full) is 0.147 on trust and runs from 0.830 to 1.049 on the other six. This is a ratio of variances and no percentage of variance explained. It exceeds one on patience Q43 because the own and cross parts covary negatively. The comparison sets trust against six surfaces, so it is no graded relation and we claim no dose–response. No bank label depends on a second dimension. In both banks all 27,636 country-pair training rows follow the sign of the pair's own dimension, with 0 mismatches in the first bank and 0 in the second. Therefore, the labels do not carry the cross talk. We claim no cause of the loss, which would need component-level analysis and interventions.

Retraining noise is small beside these effects: an unseeded retrain of one profile moves every strict surface by at most 0.112 of a standard deviation across profiles (R; online supplement). This is a narrow observed comparison of one retrained unit with its original. We did not measure variation across units.[1]

[1]The near-zero filter (N) is not promoted under the predeclared rule, and two refits support no claim of a reversal across countries. The online supplement gives the margin in full.

5.4 Instrument to respondents: the retrospective screen

Retrospectively, the screen rejects its registered null (H). The probe selected 56 of the 223 pool items under the primary rule. Their mean signed agreement with the Wave 7 country differences is $A = 0.061$ (Figure 2, panel A), with a country-bootstrap interval of $[-0.007, 0.126]$. The permutation null reassigns the same dimension and sign pairs to items drawn at random from the pool. Its distribution has mean 0.012 and ninety-fifth percentile 0.046, and the one-sided p is 0.010 (panel B).

By the registered reading, the signal is weak. Two WVS waves agree with each other on the selected items fielded in both waves with a mean Spearman correlation of 0.746 (median 0.767, 34 items). The screen's A is 8.1% of that ceiling, well below the registered bar of one third, which would be 0.249. The interval reaches zero while the permutation test rejects the null, because the two answer different questions. The test asks whether the assignments beat random assignments of the same pairs to items from the same pool. The interval asks how much A varies across samples of countries. The direction of the finding is clearer than its size. The registered reliability checks point the same way. The secondary rule gives $A = 0.070$ over 185 items with a one-sided permutation $p < 0.001$, and dropping one country at a time moves the primary A only between 0.051 and 0.072.

The per-dimension readings show where the signal sits (Table 2). Under the primary rule, only the items assigned to risk taking have an interval above zero. Their $A_d = 0.158$ [0.015, 0.290] on 13 items, of which 92.3% agree in sign. The intervals for trust and altruism include zero. The items assigned to positive reciprocity point the wrong way: their point estimate is negative, and only 18.2% of the 11 items agree in

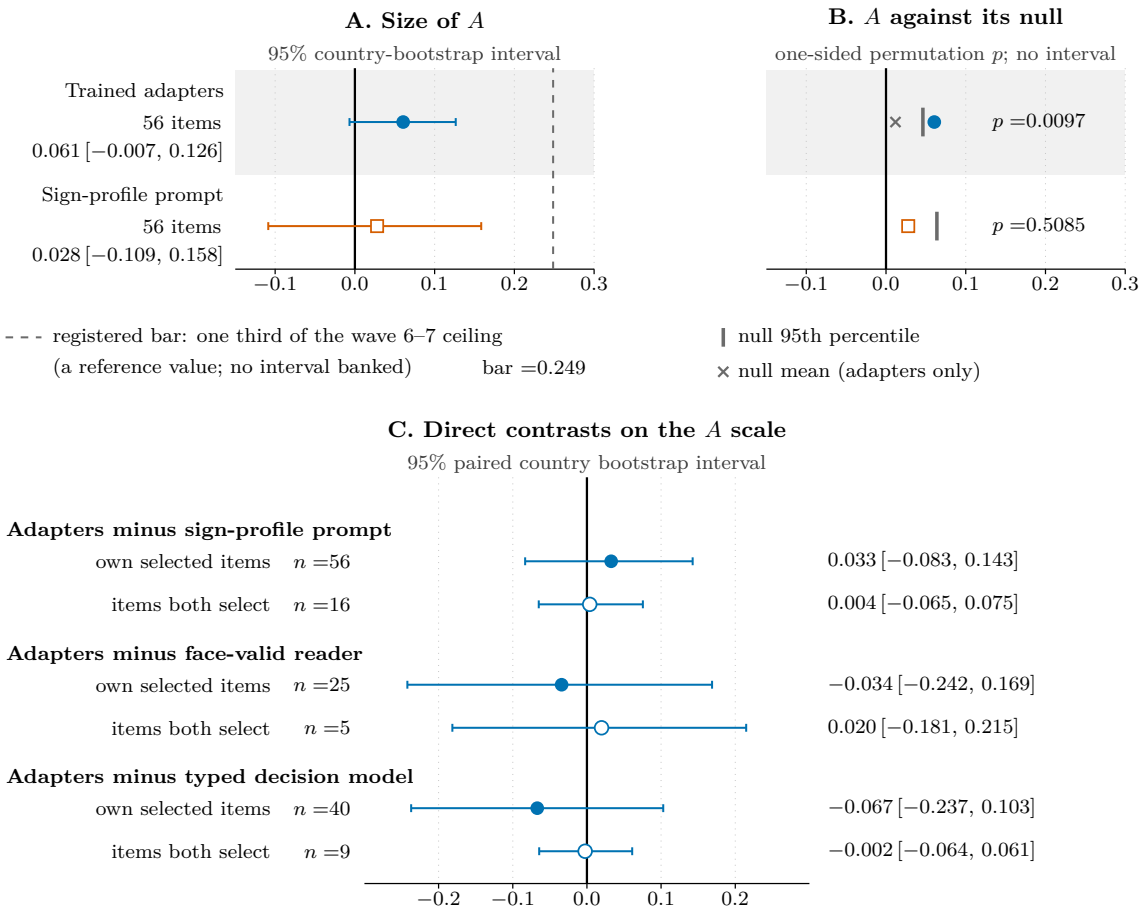


Figure 2: The retrospective held-out screen (surface H; preregistered, retrospective criterion). A is the mean signed agreement, over the selected items, between the direction the probe assigned to an item and the WVS Wave 7 country differences on that item. (A) A for the 56 items the adapters selected and for the sign-profile prompt's own selection, with 95% country-bootstrap intervals over 42 countries; the dashed line is the registered bar of one third of the two-wave human ceiling, a reference value with no interval. (B) The same A against its permutation null: the bar is the null's ninety-fifth percentile, the cross its mean (adapters only; none is banked for the prompt), and the one-sided p is printed. (C) Paired contrasts, adapters minus each comparator, on each route's own items and on items both select (n printed), with 95% paired country-bootstrap intervals; the registered primary contrasts are those with questionnaire readers on shared items. Interval and permutation p answer different questions, and an interval covering zero is not evidence of equivalence.

sign. Under the exploratory Bonferroni adjustment for four dimensions, the interval for risk taking becomes [−0.026, 0.326] and includes zero.[2]

Table 2: The second-bank check and the retrospective screen, by imprint dimension. The two checks are reported side by side and never averaged.

	S2: second bank	H: primary rule	H: primary rule, Bonferroni	H: secondary rule
Dimension	ρ [95% interval]	A_d [95% interval], items	A_d [adjusted interval]	A_d [95% interval], items
Trust	0.803 [0.615, 0.876]	0.062 [−0.047, 0.173], 12	[−0.088, 0.199]	0.092 [−0.019, 0.199], 44
Risk taking	0.210 [−0.199, 0.584]	0.158 [0.015, 0.290], 13	[−0.026, 0.326]	0.143 [0.045, 0.229], 50
Positive reciprocity	0.366 [−0.068, 0.652]	−0.071 [−0.281, 0.154], 11	[−0.332, 0.205]	−0.049 [−0.198, 0.117], 22
Altruism	0.364 [−0.057, 0.653]	0.069 [−0.036, 0.177], 20	[−0.058, 0.198]	0.040 [−0.038, 0.117], 69

Note: S2 is the Spearman correlation between second-bank adapter composites and the GPS coordinate over 42 countries, with a profile-block bootstrap (2,000 draws, seed 20261011); model side only. H is the retrospective screen against WVS Wave 7 country means: A_d is the mean signed item agreement over the selected items assigned to the dimension, with the registered country bootstrap (2,000 resamples, seed 20261015). The registered intervals are unadjusted; the Bonferroni intervals over the 4 imprint dimensions are exploratory and read off the same draws. The secondary rule is a registered reliability check. No dimension has an interval above zero on both S2 and H.

The two checks disagree by dimension. Trust survives the second bank, and its screen interval includes zero. Risk taking is the one dimension whose screen interval excludes zero before adjustment, and it is one of the three that lose their interval on the second bank. No dimension passes both registered checks, and the per-dimension cells are small.

[2]The probe's per-dimension permutation p values were not registered. For altruism under the primary rule they point the other way from the registered interval, so we report the registered intervals only.

5.5 Instrument to respondents: questionnaire readers and the prompt route

The registered contrasts do not distinguish the adapters from a careful reader of the questionnaire (Figure 2, panel C; the online supplement tabulates every contrast). Every interval covers zero, which the preregistration reads as "the adapters match a careful reader". That reading rests on few items. The coverage-matched contrasts use the 5 items that the adapters and the face-valid rule both assign and the 9 items they share with the typed decision model. The routes agree on dimension and sign for 1 and 7 of them. An interval that covers zero on so few items is consistent with a match and does not establish one. Both readers concentrate on trust, where their own-item A exceeds the 0.062 of the adapters' own trust items, but their per-dimension cells are too small to be informative.

Prompting the base model with the same six signs does not pass the registered null. The same probe applied to the 64 prompted profiles gives $A = 0.028$ [-0.109, 0.158] with a one-sided permutation p of 0.509, and it assigns its selected items to a single dimension, trust. The adapter route passes the null and the prompt route does not, but the direct difference between them is unresolved. Adapter minus prompt is 0.033 on each route's own items and 0.004 on the 16 items both select, and both intervals cover zero. The two routes agree on dimension and sign for 6 of those shared items. We therefore cannot credit the trained weights over the prompt on the screen.

Item-level assignment is stable under option order and paraphrase, whereas membership in the particular selected set churns. The registered statistic is the number of selected items whose dimension and sign are unchanged. It is 39 of 56 with reversed options, 39 with the first paraphrase and 30 with the second. No variant changes more than half, so the registered condition for wording dependence is not met, although the

second paraphrase comes close. Membership is another matter: the primary rule is a top-quartile cut on $|t|$, and only 18, 29 and 16 of the 56 items are selected under both the original readout and each variant. Dimension and direction are more stable than the list of top-ranked items.

5.6 Instrument to respondents: agreement with the human composite

The next two subsections report descriptive follow-up analyses (D), added after the registered results had been read. None is preregistered and none enters a registered family. They set the instrument beside the respondents of the Wave 7 file on the strict battery, and every statement is retrospective. Figure 3 gives three Spearman correlations over the 42 countries for each of seven surfaces. They pair the adapter composite with the GPS, the adapter composite with the human composite of the same items, and the human composite with the GPS. The panels have different statistical targets and bootstrap units, so their widths are not comparable.

On trust, the first-cohort adapters track the GPS at 0.744 [0.500, 0.865]. They do not detectably reach the human composite, at 0.295 [−0.044, 0.547]. All seven adapter–human intervals include zero, which is not evidence that no association exists. The largest point estimate belongs to negative reciprocity, 0.389 [−0.008, 0.694], a surface on which the adapters do not track the GPS. The second-cohort adapters give the same picture on trust, with an adapter–human correlation of 0.340 [−0.025, 0.586]. Positive reciprocity is the one second-cohort interval that excludes zero, and only just (online supplement).

The link between the GPS and the human composite is itself loose on these items. Its interval excludes zero on four of the seven surfaces. Among them, it is positive for

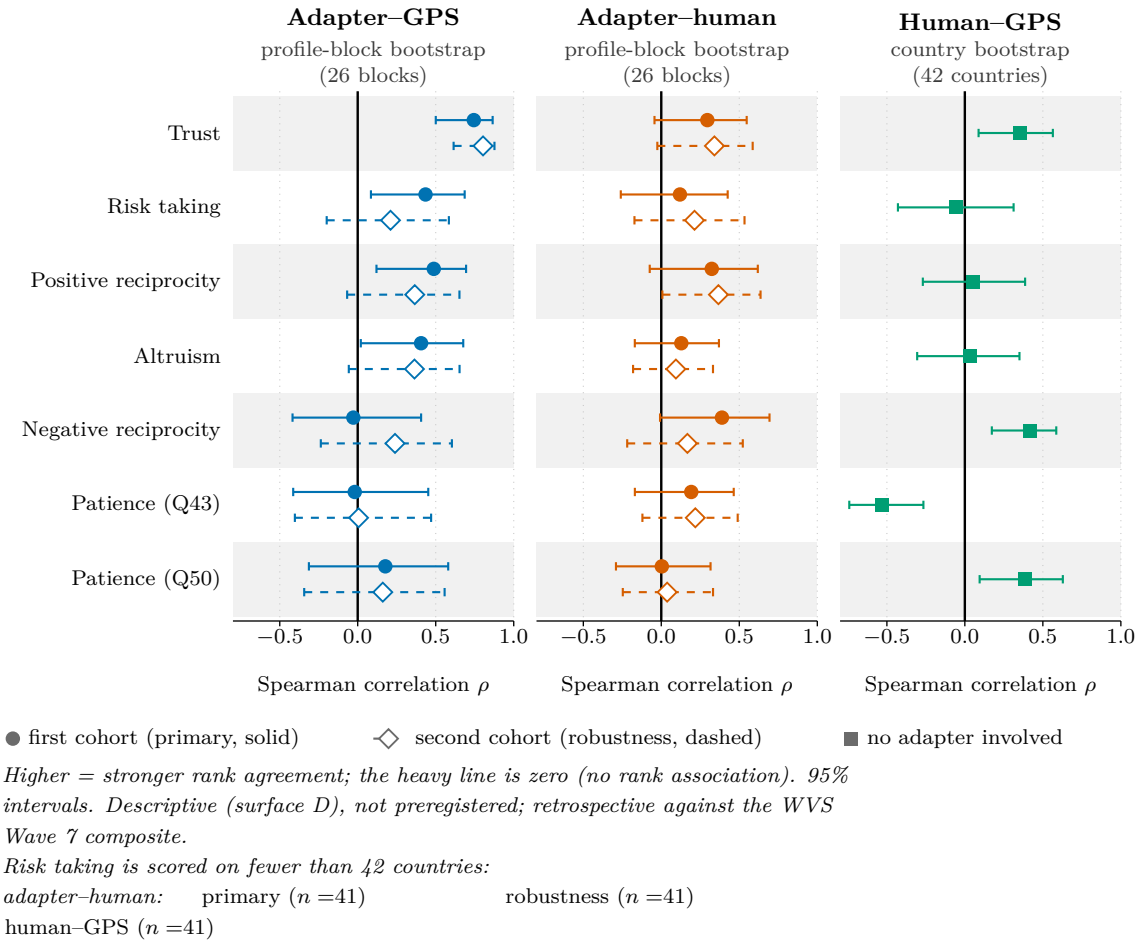


Figure 3: The three links on seven strict-battery surfaces (surface D; descriptive, not preregistered; retrospective). Each mark is a Spearman correlation across countries with a 95% interval; the heavy line is zero. Left: first-cohort adapter composite with the GPS coordinate of the surface's own dimension. Middle: adapter composite with the Wave 7 composite of the same items. Right: that human composite with the GPS. Circles are first-cohort adapters (primary), diamonds second-cohort adapters (robustness), squares involve no adapter. Intervals in the adapter panels come from a bootstrap over the 26 realized sign profiles, those in the right panel from a bootstrap over the 42 countries. Risk taking covers 41 countries in the two human columns.

trust at 0.351 [0.088, 0.564] and negative for patience Q43 at −0.533 [−0.741, −0.266]. We read this column as a reference for scale. It is no upper bound on what an adapter could reach, because it does not bound a scorer that is nonlinear or constant within sign profiles. Nor is it evidence that the anchor limits the instrument. Together, the three panels say that the adapters carry the GPS order on trust and that the GPS order is only loosely related to the human composite. They also show that the adapters' order of countries does not detectably agree with the human composite on any surface. The exact values for every cell are in the online supplement.

5.7 Instrument to respondents: order versus distance

The correlations above concern the order of countries. A different question is how far the instrument's response distribution lies from the survey's. We measure it by the country-equal mean total variation distance (TVD; lower is closer; D) in four cells. They are the base model and the first-cohort adapters, each with and without the country-named persona prompt.

The base model's mean distance is 0.470. The adapters sit farther from the survey at 0.515, and the persona prompt sits closer at 0.365 (panel B of Figure 4). A cell mean carries no interval, whereas the paired contrasts of panel C each carry one. Relative to the base model, the adapters are farther by 0.045 [0.006, 0.093] under profile blocks, and the persona prompt is closer by −0.105 [−0.122, −0.086]. Country resampling gives the same readings. With both interventions the distance is 0.420. Adding the persona prompt to the adapters moves the distance by −0.096 [−0.115, −0.078], and adding the adapters to the persona prompt moves it by 0.055 [0.016, 0.097]. The interaction contrast is the adapters' effect with the persona prompt minus

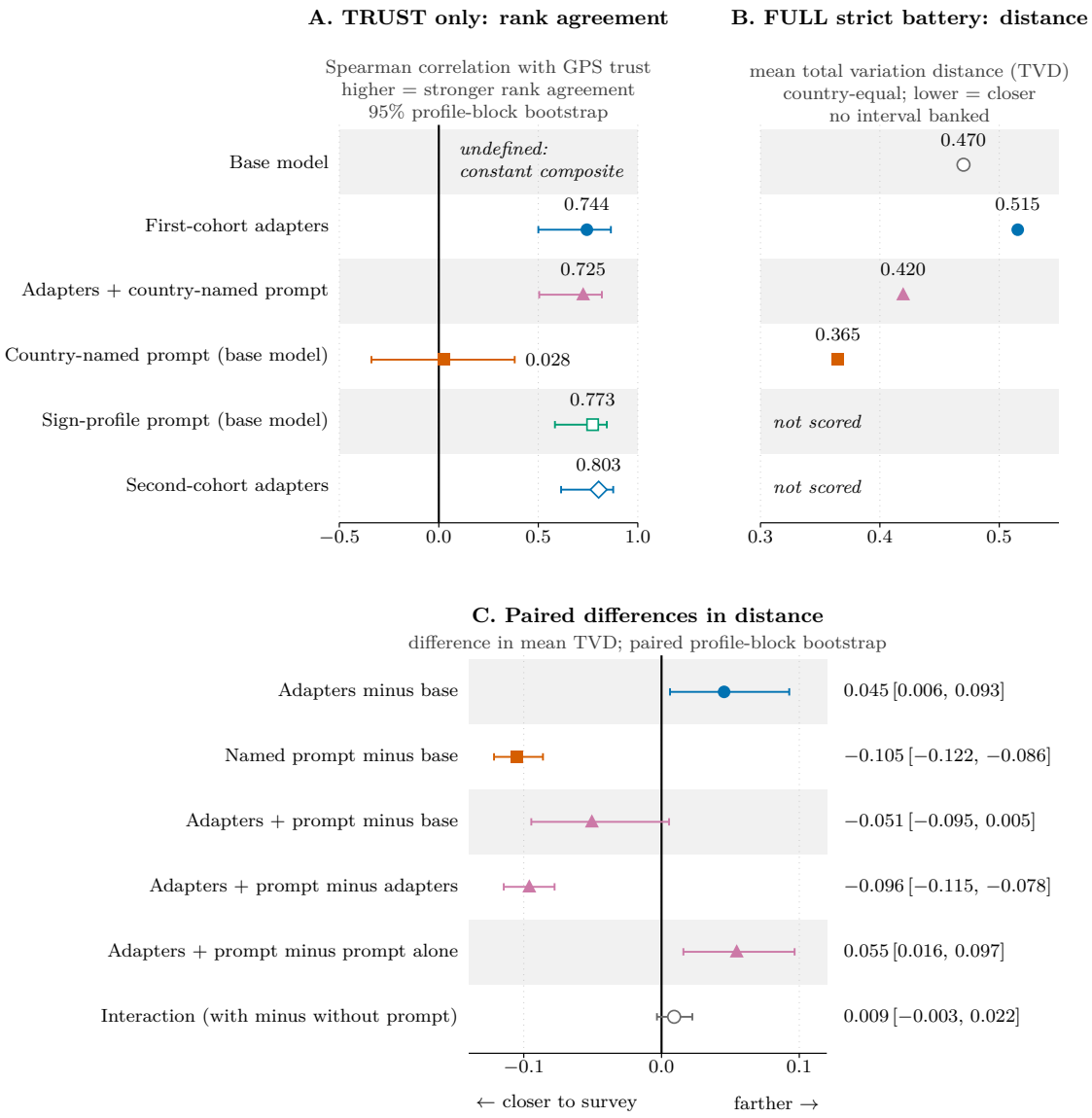
their effect without it. Under profile blocks it is 0.009 [−0.003, 0.022], an interval that includes zero. Therefore, the data show no detectable departure from additivity.

The ordering of the cells by distance does not follow their ordering by trust agreement with the GPS. The persona prompt has the lowest distance and no trust order (Section 5.2). The adapters have the highest distance and keep the trust order. Adding the persona prompt to the adapters keeps the trust order at 0.725 [0.505, 0.820] while the distance falls.

6 Discussion

We asked a language model only about the group, and the answer comes in two parts that point in opposite directions. Conditioned on declared group signs, the instrument keeps the GPS order of countries on trust across both scenario banks. Checked against the Wave 7 respondents, its order of countries does not detectably agree with theirs, and its response distributions sit farther from theirs than the base model's do. In other words, it can order countries the way a cross-national survey does while answering unlike the people in them.

The conditioning method: what the signs carry. The declared sign carries the trust order, whichever channel supplies it. The adapters keep the trust order across scenario banks (Section 5.2). A country name does not recover that order, whereas a prompt that states the six signs does. On trust the sign prompt correlates with the GPS at 0.773 [0.584, 0.844], against 0.028 [−0.339, 0.381] for the named country. In the screen, we could not resolve the contrast between the adapters and the sign prompt. Therefore, the evidence cannot credit the trained weights over a prompt, and the statement concerns trust only. The decomposition of the factorial shows where



Descriptive (surface D), not preregistered. The base model's trust–GPS correlation is undefined (constant composite), not zero. Trust is one dimension; TVD scores the full strict battery. The crossing of the two readouts is descriptive: not a causal effect of fine-tuning, nor a general law about weights versus prompts.

Figure 4: Order and distance for the same routes (a descriptive display, not preregistered; the distances are retrospective). (A) Trust only: Spearman correlation of each route's trust composite with the GPS trust coordinate over 42 countries, with 95% profile-block intervals. The first-cohort, second-bank and country-named values are those of Section 5.2, where their own status is stated. (B) Mean total variation distance (TVD) to the survey answer shares over the full strict battery, country-equal; routes not scored on this battery are marked. (C) Paired differences in mean TVD with 95% profile-block intervals; negative means the first-named cell is closer. Panel A scores trust only and panels B and C all dimensions, so they are different metrics.

trust differs from the other surfaces. Every own effect is positive, so the own part of the main-effects prediction restates the correlation between the signs and the GPS on every surface. What differs is how much the other five signs add. On trust they add little, with Var(cross)/Var(full) at 0.147. On the other six surfaces they carry most of the variation, with ratios from 0.830 to 1.049. The decomposition describes behavior and leaves open why the other dimensions lose their order.

Training on sign-labeled pairs keeps the direction of each declared input and does not deliver separate dimensions. Each sign moves its own surface in the intended direction and also moves other surfaces. One common direction carries about half of the variance across surfaces, and the planned separability check fails on both routes. Prompting with the same signs is more entangled still, and three of its own effects point the wrong way. Entanglement under intervention has also been reported when cultural values are edited through a model's internal activations (Dang and Masud, 2026). Our factorial adds a joint measurement for trained adapters and for a prompt that carry the same declared information. We make no claim about the best training recipe.

The evaluation protocol: what it separates. Each part of the protocol separated something that a single agreement figure would have merged. Notably, separating the three links showed that the GPS's own agreement with the human composite is loose on these items. Weak adapter–human agreement therefore cannot be attributed to the instrument alone. It could arise from the instrument, from the mapping between items and the GPS construct, or from limited coherence between the two surveys. The design cannot tell these apart. The registered screen and its freeze sequence are fixed before the human criterion is read, and we label the descriptive analyses added

afterwards as such. Another team can therefore apply the protocol to a different conditioning method, anchor or criterion.

Order and distribution. The split between order and distribution is what a group-level design gives us reason to expect. We chose to model the group directly and left the aggregation over people to the GPS. An order of countries needs only a score that rises with the group's latent preference, and no representative respondent whose answers equal the population's (Kirman, 1992). A single choice policy per country, however, has no spread within the country built in. That is, it can carry an order and has no reason to match the distribution of respondents' answers. We offer one reading without testing it: the training signal states which of two responses is preferred and says nothing about how strongly, or about how widely respondents spread across the options. The order of countries is a quantity that signs speak to, whereas a distance to the survey's response distributions is one they carry no information about. This argument explains why the adapters have no reason to approach the survey's response distributions. It does not explain why they move away from them, or why a country-named prompt moves toward them.

The evidence shows the two targets crossing (Section 5.7). The adapters keep the anchor's order of countries on trust and lie farther from the survey's response distributions than the base model, by 0.045 [0.006, 0.093] in total variation distance. A country-named prompt lies closer, by -0.105 [-0.122, -0.086], and carries no trust order. If agreement were one quantity, the routes would fall in one ranking. They fall in two. The contrast describes one anchor, one instrument family and one survey, and it does not show that the country-named prompt matches respondents. A country name has been found to recover a country ranking in the European Social Survey while supporting no distributional inference (Wang, 2026). The separation of group-level

order from distributional fidelity is established (da Silva et al., 2026). Our case adds one method whose two answers point in opposite directions, so order and distribution have to be validated separately.

What a prospective validation would have to establish. A prospective validation would have to establish three things that the retrospective design cannot. First, the probe's selections and a human-only rule for choosing items would both be declared before any new items are fielded. Second, the selected items would have to separate countries in the assigned direction in new data. They would also have to do so better than the human-only rule on matched countries and items. Third, the dimension on which the signal appears would have to be one that the second bank also supports. We would read a result of that kind as licensing the use before fielding. Whether such a test would find the signal on trust, the one dimension the second bank supports, is the open question this paper leaves.

Limits. Five limits apply to every result above. First, the human criterion is retrospective and is the only human source. Registration and the freeze guard protect against tuning the probe to it. They cannot make it independent of the survey program from which the item pool was drawn. Second, we chose the country as the unit, and our claims stay at that unit. The instrument is built from signs and cannot represent magnitudes, differences among countries that share a sign profile, or variation among people within a country. Country-level agreement does not establish individual-level validity or measurement invariance (Davidov et al., 2014; Bratt, 2025). Third, all our adapters share one base model, one training recipe and one readout convention. Their code probabilities have a dispersion that need not match human response distributions. Fourth, the per-dimension cells are small and the selected set churns under readout

variants. Fifth, the noise checks are narrow. One unseeded retrain is a narrow observed comparison for one unit, and two near-zero refits support no claim across countries.

Intended use. Within those limits, the instrument has a modest and inspectable use. A survey designer can read the dimension and direction it assigns to candidate items as one input beside expert review and human pretesting, keeping in view that no dimension passed both checks. We see it as a complement to fielding and pretesting, and the evidence gives no reason to replace them. In the terms of Broska et al. (2025), its readings are potentially informative observations that still require a human criterion.

7 Conclusion

Asked only about the group, a language model can order countries the way a benchmark cross-national survey does while answering unlike the respondents of a second survey. We built synthetic cultural agents that compile declared aggregate preference moments into choice policies and read survey items without naming a group. We then tested each link of the chain from anchor to instrument to respondents separately.

Between anchor and instrument, the method holds in a limited sense. Training installs the synthetic labels on pairs the adapters never saw, and on trust the adapters keep the anchor's order of countries across a change of scenario bank. A prompt stating the same signs keeps that order too, whereas a country name does not. That is, the declared signs carry the trust order, and the evidence cannot credit the trained weights over a prompt. The adapters do not keep the declared dimensions separate.

Between instrument and respondents, the evidence is weak. A preregistered screen of held-out items, read against a retrospective criterion, gives a small signal. The

registered comparisons do not separate it from careful readers of the questionnaire, and no dimension passes both the second-bank check and the screen. Descriptively, the adapters' order of countries does not detectably agree with the respondents' composite. Their response distributions lie farther from the survey than the base model's, whereas a country-named prompt lies closer with no trust order. Keeping an order of countries and matching respondents' response distributions are different targets, and a design built at the level of the group gives a reason to expect that they come apart.

We offer the method as an inspectable, group-level reading of candidate items that a researcher may use beside human pretesting, as a complement and never in its place. Our evaluation protocol applies to other conditioning methods and other anchors. The test that would license more is prospective: probe-selected items fielded to people after the selection and a human-only comparison rule have both been declared. Until that test is run, pre-piloting remains untested.

Preregistration Statement

The held-out screen (Sections 4.3 and 5.4) was preregistered. The registration document was committed to the project repository and timestamped by the git tag `prereg-heldout-2026-10-14`, pushed before any held-out item was scored by an adapter. Amendment 1 (repair of 12 items whose standalone wording could not be answered) was made after the first model-side probe pass had been computed and read, and before any human value was read. The freeze file was committed, and Amendment 2 (removal of those items from both comparator routes) was decided, before the single read of held-out human data; the text of Amendment 2 was written afterwards and records that earlier decision. The criterion data, WVS Wave 7, are public and existed before the study, so the registration fixes the analysis before the

authors read the human values and the criterion remains retrospective. All registered analyses of the screen are reported in Section 5.4, Table 2 and Figure 2, except two descriptives (item-level agreement after dropping items with small $|a_i|$, and the face-valid reader's statistic on all of its items including negative reciprocity), which are reported in the online supplement. The Bonferroni intervals, the wave-specific screen statistics, the scale-free factorial views and the factorial route contrast are exploratory. The factorial's separability check (V5) was fixed in the project's analysis plan, separately from the registration. The second-bank decision rule was fixed before the second cohort was trained and was not part of the registration. An anonymized copy of the registration is provided with the submission.

Code, Materials, and Data Availability

Analysis code, the item pool and its overrides, the frozen selection, the comparator files, all model scores, and the scripts that build every number in this article from those scores are available for review at an anonymized link. Model adapters and score archives are released at acceptance. The frozen inputs are identified by SHA-256 hashes: the amended item pool `19241f6f103c51d6`... as written to disk, whose line-ending-normalized form in the repository is `a1dbceea1a33da05`...; the overrides file `0f5cd2bbd88ef6ea`...; and the frozen selection `782b7d2123f9758e`.... The pool is regenerated byte for byte from the overrides file by the pool-building script; full hashes and the line-ending convention are in the online supplement. WVS and GPS data are available from their providers (Haerpfer et al., 2022; World Values Survey Association, 2022; Falk et al., 2018) and are not redistributed; no respondent-level data or country-level human means are included in the replication materials.

Institutional Review

This study analyzes publicly available, de-identified secondary survey data and involves no interaction with human participants.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Generative AI Use

Large language models are the object of study in this article. In addition, an AI coding and writing assistant was used under the authors' direction to write analysis code and to draft and edit manuscript text; the authors reviewed all code, numbers and text and take responsibility for them.

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